



</> PROGRAMMING FUNDAMENTALS

Session - 4

Organization : TeachToTech

Instructor : Ayush Raj

Duration : 1.5 hr

(knight @leetcode, 4 ★ @codechef, Specialist @codeforces)

Why Do We Need Loops?

01

The Problem

Printing "Hello" 5 times? Copy-paste
printf works.

02

The Real Problem

What if you need to print it 10,000
times?

03

The Solution

Loops repeat code blocks until a condition is met.

for

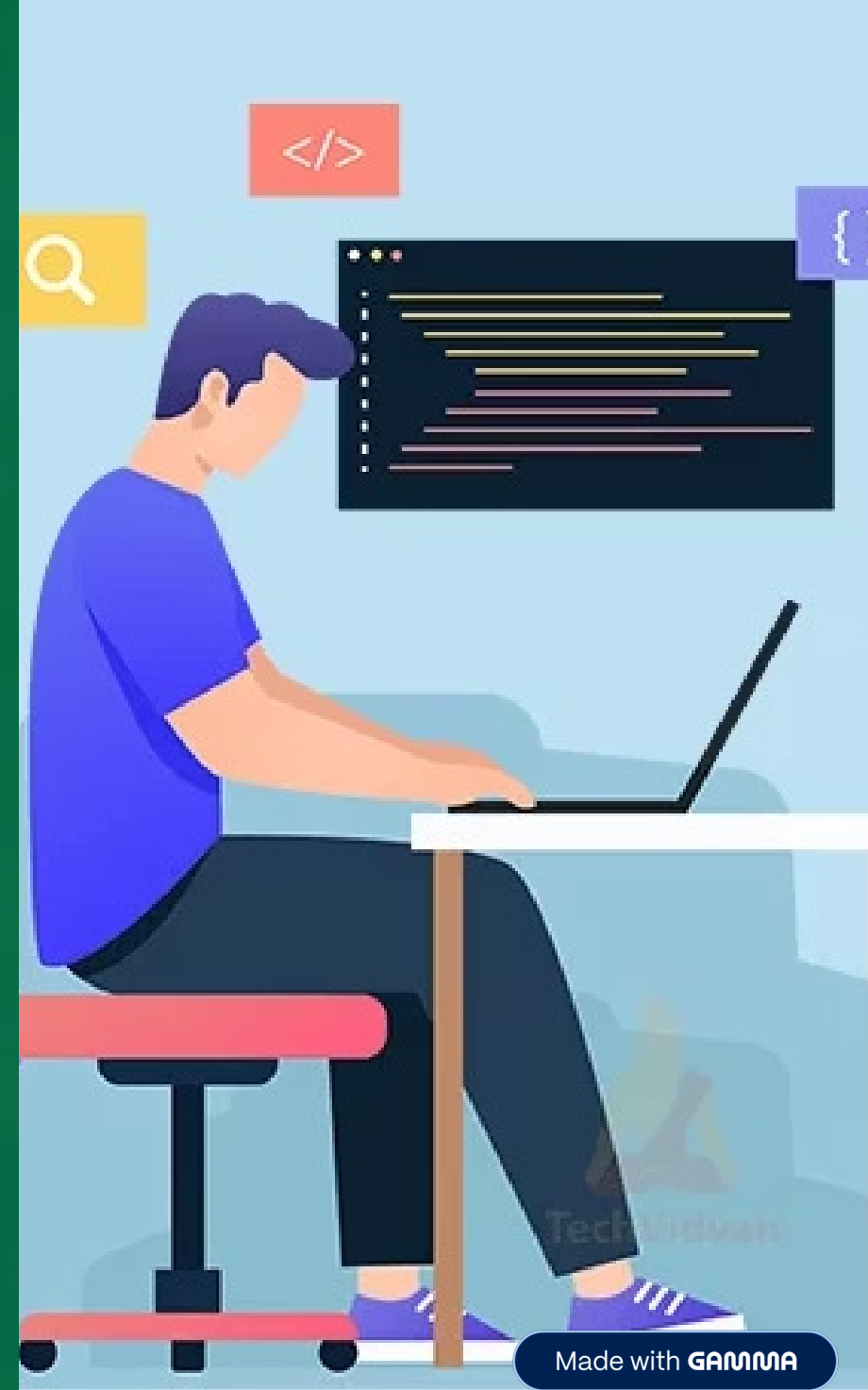
When you know how many
times

while

When you check condition first

do-while

When you must run at least once



The Syntax Comparison

for Loop

```
for(start; stop; step) {  
  ...  
}
```

Best when iterations are known

while Loop

```
while(condition) {  
  ...  
}
```

Check condition first

do-while Loop

```
do {  
  ...  
} while(condition);
```

Runs before checking

📌 **Key Difference:** do-while runs the code before checking the condition.

[illegible]

Basic Activity - 1 to N

Task: Print numbers from 1 to 10

```
int i;
for(i = 1; i <= 10; i++) {
    printf("%d ", i);
}
```



For Loop

Best used when you know the number of iterations in advance.

```
for (init; condition; inc) {
    // code
}
```



While Loop

Best when iterations are unknown
and depend on a condition.

```
while (condition) {  
    // code  
}
```



Do-While Loop

Executes code block at least once before checking condition.

```
do {  
    // code  
} while (condition);
```

Loops (Problem Set - 1)



LeetCode



Palindrome Number ...

Can you solve this real
interview question?...



LeetCode



Single Number – Leet...

Can you solve this real
interview question? Singl...



LeetCode



Valid Anagram – Leet...

Can you solve this real
interview question? Valid...



LeetCode



Missing Number – Le...

Can you solve this real
interview question?...

Loops (Problem Set - 2)



LeetCode



Find the Difference - ...

Can you solve this real interview question? Find...



LeetCode



Fibonacci Number - L...

Can you solve this real interview question?...



LeetCode



Reverse Integer - Lee...

Can you solve this real interview question?...



LeetCode



Missing Number - Le...

Can you solve this real interview question?...

Problem Solving: Key Algorithms

- **Reverse Integer / Palindrome**

LeetCode 7 & 9: Reverse digits of number x

```
int reversed = 0;
while (x != 0) {
    int digit = x % 10;
    reversed = reversed * 10 + digit;
    x = x / 10;
}
```

- **Single Number**

LeetCode 136: Find non-duplicate using XOR

$$a \oplus a = 0, a \oplus 0 = a$$

- **Valid Anagram**

LeetCode 242: Check if strings use same characters

Use frequency counter: `count[26]` array

- **Missing Number**

LeetCode 268: Find missing in range [0, n]

Expected Sum: $\frac{n \times (n+1)}{2}$

Nested Loop Discussion :

Syntax :

```
for ( initialization; condition; updation) {  
  
    for (initialization; condition; updation) {  
  
        ....  
  
        statements  
  
    }  
  
}
```

Lab Activity: Pattern Printing

Write nested loops to print this pyramid:

```
*  
* *  
* * *  
* * * *
```

```
for(int i=1; i<=4; i++) {  
    for(int j=1; j<=i; j++) {  
        printf("* ");  
    }  
    printf("\n");  
}
```

A dense, vibrant background of various tropical leaves, including large monstera leaves and palm fronds, in shades of green and yellow.

THANK
YOU!

Happy Coding....